

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

27th May, 2014

Subject SA6 – Investment

Time allowed: Three hours (14.45* - 18.00 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which are meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the Candidate should answer the question with reference to Indian environment.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** You work with a multinational infrastructure company (ABC Co). The company, which is listed on NSE, has 1 million shares in issue and the current ex div ordinary share price is Rs 250 per share. The company also had issued bonds in the past with a book value of Rs 60 million and their current ex interest market price is Rs 104 per Rs 100 bond. The current after-tax cost of debt of ABC Co is 7% and the tax rate is 30%.

The recent dividends per share of the company are as follows.

Year	2009	2010	2011	2012	2013
Dividend per share (Rs)	19.38	20.20	20.41	21.02	21.80

- 1) One of the directors has proposed to decrease the weighted average cost of capital of the company and hence increase its market value, by issuing 40 million Rupees of bonds at their par value of Rs 100 per bond. These bonds would pay annual interest of 8% before tax and would be redeemed at a 5% premium to par after 10 years. You can also use the information on dividends paid by the company and other information given in the question
 - i) Calculate the after-tax weighted average cost of capital and the market value of ABC Co in the following circumstances:
 - a) before the new issue of bonds takes place
 - b) after the new issue of bonds takes place (8)
 - ii) Discuss the director's view that issuing traded bonds will decrease the weighted average cost of capital of ABC Co and thereby increase the market value of the company. (3)
 - iii) Discuss why your company may choose convertible debt over a bank loan of a similar maturity as a source of finance. (3)
- 2) The treasury team in the company wants your help in understanding various portfolio management issues and models explaining risk return trade-offs.
 - i) Please describe Capital Asset Pricing Model with background to earlier formulations of portfolio theory such as the mean-variance and capital market line methods. (3)
 - ii) What are the practical advantages and disadvantages of using the Capital Asset Pricing Model? (3)
- 3) The company has also asked you to compare 'passive' fund management and 'active' fund management. Please show the style comparison and then also explain, which style – passive or active, is more suitable for the Indian investment market? (8)
- 4) The company makes a lot of investment in different projects and wants your help in incorporating risk into the investment appraisal process. Please suggest, the methodologies to allow for risk while doing investment appraisal of a project. (3)

- 5) The company's investment portfolio comprises of two assets with the following expected returns and standard deviations:

	Asset 1	Asset 2
Expected Return	7.5%	12%
Standard Deviation	11%	21%

- i) Calculate:

- a) The expected return for a portfolio which is equally weighted between the two assets. (2)
- b) The correlation coefficient for the two-asset portfolio, assuming that the covariance is 27. (1)
- c) The variance of returns for the equally weighted portfolio, assuming a covariance of 27. (2)
- d) The standard deviation of returns for the equally weighted portfolio. (2)

- ii) Comment on why the portfolios with two or more assets have lower risk than the combined risk obtained from the weighted arithmetic average of the asset risks. (1)

- iii) In principle, the two-asset model of portfolio risk can be applied to portfolios which include far greater numbers of assets. Explain how this approach helps in the understanding of total risk in such portfolios and discuss the practical problems of applying the model in this fashion. (2)

- iv) A portfolio consists of two assets which have a correlation coefficient of 0.8. Another two-asset portfolio has a correlation coefficient of -0.8. In the absence of any other information, which portfolio should you invest in? Justify your reasoning. (1)

- 6) The company management feels that the corporate bond market in India has huge potential for development and the treasury team should look beyond the government securities. Highlight the recent changes and challenges in the Indian bond market with reference to corporate bonds. (4)

- 7) Identify and discuss briefly the factors that influence the market value of traded bonds. (3)

- 8) State the features of a bond index. (3)

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- Q. 2) With reference to Credit Default Swaps (CDS), answer the following:

- i) What is a credit event? State with examples. (2)

- ii) Explain CDS (2)

- iii) Demonstrate using arbitrage, how the price of CDS is related to the yield of the underlying bond. (4)

- iv) Highlight the main applications and constraints of CDS. (4)
 - v) What are the risks in CDS for a protection buyer? (3)
 - vi) What should be covered under a risk management framework for CDS? (3)
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- Q. 3)** You are a qualified investment consultant in India offering investment advice to various organisations and also to high net-worth individuals (HNIs). You are currently working on two assignments. One of the clients is a Japanese company which wants to make investments in India and seeks your advice on accessing the Indian market.
- i) Briefly describe the macro-economic environment in India with a focus on how conducive it may or may not be for the Japanese company to invest in the present circumstances. (5)
 - ii) Please discuss the following investment avenues available to foreign investors for making investment in India
 - a) Investment made as Foreign Direct Investment (FDI)
 - b) Foreign Institutional Investor route (FII)
 - c) Qualified Foreign Investors (QFI)
 - d) American Depository Receipts (ADRs)/ Global Depository Receipts (GDRs)
 - e) Foreign Currency Convertible bonds (FCCBs)/ Foreign Currency Exchangeable bonds (FCEBs) (10)
 - iii) Another client, on whose assignment you are currently working is a salaried professional and is 35 years of age. Her current gross annual income is over INR 5,000,000. You gather from your discussions that she has a balanced approach towards taking risk. Her housing loan has a unique feature called “Maximum Advantage” which makes the housing loan work like an overdraft account (OD account). If she parks any liquid cash available with her in that loan account the outstanding loan amount reduces and for that period the interest is charged only on such reduced loan value. Thus putting any amount of liquid money in that account is as good as repayment of loan up to that amount but with the additional flexibility of future withdrawals from that account whenever liquidity is required.

Your client’s balance sheet as on 31st March 2014 was as follows:

Liabilities and Capital	Amount (INR million)	Assets	Amount (INR million)
Capital	7.0	House	7.5
Loans		2 Cars	1.2
Education loan @ 14% p.a.	1.5	Provident Fund	1.1
Housing loan @ 12% p.a.	5.0	MF Savings (Equity)	1.0
Car Loan @ 13% p.a.	1.5	Insurance (Endowment)	0.2
Current Liabilities	0.0	Cash and Bank balances	4.0
Total	15.0		15.0

- a) Discuss whether the split between capital and loans availed looks reasonable and state with reasons whether the client should continue with her loans? (4)
 - b) Discuss the key features of the assets and state with reasons whether the asset mix currently held is reasonable. (4)
 - c) Evaluate the liabilities and assets together to see whether any restructuring of the balance sheet (by changing the assets and/or liabilities) may help in increasing the returns with similar levels of risks and/or reduce the risks with similar level of returns. (7)
 - iv) A privately owned insurance company in India is considering an initial public offering. You know that a large listed conglomerate owns 75% of this insurance company. Your HNI client in the question above is eager to get exposure in this company before it is listed. Describe how a synthetic share can be constructed of this unlisted company. (5)
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